RAJALAKSHMI ENGINEERING COLLEGE

Prathesha S 251801238-2025-AIDS

P2

Started on	Thursday, 6 August 2026, 10:11 AM
State	Finished
Completed on	Thursday, 6 August 2026, 10:15 AM
Time taken	3 mins 52 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

void function(int n) {
    int count = 0;
    int i = 1; count++;
    int s = 1; count++;
    while(s <= n) {
        count++; // For the true loop condition evaluation

        i++;     count++; // For loop step increment
        s += i;  count++; // For accumulated sum modification
    }
    count++; // For the final false loop condition evaluation that exits the loop

    printf("%d\n", count);
}

int main() {
    int n;
    if (scanf("%d", &n) == 1) {
        function(n);
    }
    return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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P2

Started on	Wednesday, 26 August 2026, 10:33 AM
State	Finished
Completed on	Wednesday, 26 August 2026, 10:41 AM
Time taken	7 mins 11 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int count = 0;

void func(int n)
{
    count++; // For if statement evaluation
    if (n == 1)
    {
        count++; // For printf("*")
    }
    else
    {
        int i;
        for (i = 1; i <= n; i++)
        {
            count++; // For loop test i <= n

            int j;
            for (j = 1; j <= n; j++)
            {
                count++; // For inner loop test j <= n
                count++; // For first printf("*")
                count++; // For second printf("*")
                count++; // For break statement
                break;
            }
        }
```

```
        count++; // For final outer loop test evaluation
    }
}

int main()
{
    int n;
    scanf("%d", &n);
    func(n);
    printf("%d", count);
    return 0;
}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Prathesha S 251801238-2025-AIDS

P2

Started on	Wednesday, 26 August 2026, 10:41 AM
State	Finished
Completed on	Wednesday, 26 August 2026, 11:47 AM
Time taken	1 hour 5 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int num, i;
    int count = 0;

    if (scanf("%d", &num) != 1) return 0;

    // 1. Initialization (executed 1 time)
    count++; // for i = 1

    for (i = 1; i <= num; ++i) {
        // 2. Loop condition check for each iteration + final fail check
        count++;

        // 3. If condition check
        count++; // for if (num % i == 0)

        if (num % i == 0) {
            // 4. Print execution
            count++;
        }
    }

    printf("%d", count);
    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔

	Input	Expected	Got	
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Prathesha S 251801238-2025-AIDS

P2

Started on	Wednesday, 26 August 2026, 10:49 AM
State	Finished
Completed on	Wednesday, 26 August 2026, 11:53 AM
Time taken	1 hour 3 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int n;
    int count = 0;

    if (scanf("%d", &n) != 1) return 0;

    // 1. Initialization of 'c = 0'
    count++;

    // 2. Loop structures
    for (int i = n / 2; i < n; i++) {
        count++; // Outer condition check (i < n)
        for (int j = 1; j < n; j = 2 * j) {
            count++; // Middle condition check (j < n)
            for (int k = 1; k < n; k = k * 2) {
                count++; // Inner condition check (k < n)
                count++; // c++ execution
            }
            count++; // Inner loop final false check
        }
        count++; // Middle loop final false check
    }
    count++; // Outer loop final false check

    printf("%d", count);
    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔

	Input	Expected	Got	
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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P2

Started on	Wednesday, 26 August 2026, 10:53 AM
State	Finished
Completed on	Wednesday, 26 August 2026, 11:56 AM
Time taken	1 hour 3 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int n;
    int rev = 0, remainder;
    int count = 0;

    if (scanf("%d", &n) != 1) return 0;

    count++; // rev = 0 execution

    while (count++, n != 0) { // count++ for while condition check
        count++; // remainder = n % 10
        remainder = n % 10;

        count++; // rev = rev * 10 + remainder
        rev = rev * 10 + remainder;

        count++; // n /= 10
        n /= 10;
    }

    count++; // print(rev) execution

    printf("%d", count);
    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔

	Input	Expected	Got	
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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